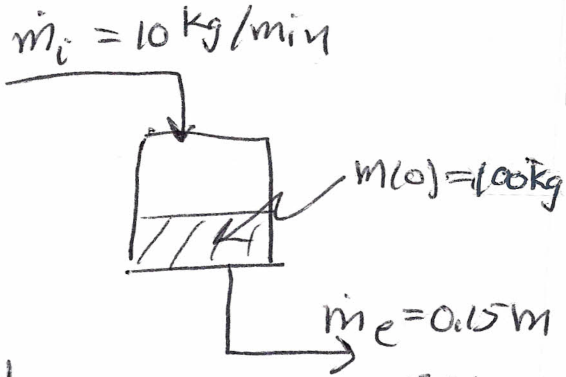
ChE 210 Spring 2020

Exam # 2



(1) (a) $\dfrac{dm}{dt} = \dot{m}_i - \dot{m}_e = 10 - 0.15m$

$$\int_{100}^{m} \frac{dm}{10 - 0.15m} = \int_0^t dt = t$$

let $u = 10 - 0.15m$
$du = -0.15\,dm$

$m [=] \mathrm{kg}$
$\dot{m}_e [=] \mathrm{kg/min}$

$$-\frac{1}{0.15}\int_{u_0}^{u} \frac{du}{u} = t$$

$$-\frac{1}{0.15}\ln\frac{u}{u_0} = -\frac{1}{0.15}\ln\frac{10 - 0.15m}{10 - 0.15(100)} = t$$

$$\frac{10 - 0.15m}{-5} = e^{-0.15t}$$

$$\boxed{m = \frac{10 + 5e^{-0.15t}}{0.15}}$$

(b) $m(t = 10\,\mathrm{min}) = \dfrac{10 + 5e^{-1.5}}{0.15} = 74.10\ \mathrm{kg}$ @ 10 min

Also $m(t \to \infty) = \dfrac{10}{0.15} = 66.\overline{66}\ \mathrm{kg}$ as $t \to \infty$

(c) $\dfrac{dm_A}{dt} = w_i \dot{m}_i - w\,\dot{m}_e$

$m_A = w\,m$
$\dot{m}_i = 10\ \mathrm{kg/min}$
$w_i = 0.1\ \mathrm{kgA/kg}$
$\dot{m}_e = 0.15\,m$

$$\frac{dm_A}{dt} = (0.1)(10)\frac{\mathrm{kgA}}{\mathrm{min}} - 0.15\,w\,m$$

$$\frac{dm_A}{dt} = 1 - 0.15\,m_A$$

$$\int_{m_{A_0}}^{m_A} \frac{dm_A}{1 - 0.15m_A} = \int_0^t dt$$

let $u = 1 - 0.15 m_A$
$du = -0.15\, dm_A$

$$-\frac{1}{0.15} \ln \frac{1 - 0.15 m_A}{1 - 0.15 m_{A0}} = t$$

$$\frac{1 - 0.15 m_A}{1 - 0.15(60)} = e^{-0.15t}$$

$m_{A0} = (0.6)(100)$
$= 60 \text{ kg A}$

$M_A(t \to \infty) = 6.\overline{6} \text{ kg A}$
$\therefore M_B(t \to \infty) = 60. \text{ kg B}$

$$m_A = \frac{1 + 8e^{-0.15t}}{0.15} \overset{t=10\text{min}}{=} 18.57 \text{ kg A}$$

$$w(t=10\text{min}) = \frac{18.57 \text{ kg A}}{74.10 \text{ kg}} = \underline{\underline{0.2506}}$$

$$w(t) = \frac{m_A}{m} = \frac{1 + 8e^{-0.15t}}{10 + 5e^{-0.15t}}$$

$$w(t = 10\text{min}) = \frac{1 + 8e^{-1.5}}{10 + 5e^{-1.5}} = \underline{\underline{0.2506}}$$

CHE 210 Exam #2
Sp 2020

(2)(a) Overall Balance (4 eqn (A,B,P,Q) and 4 unks. $A_1, B_1, \dot{\xi}_1, \dot{\xi}_2$)

4th $A: A_5 = A_1 - 2\dot{\xi}_1 \Rightarrow A_1 = A_5 + 2\dot{\xi}_1 = 10 + 2(121) = 252$

3rd $B: B_5 = B_1 - \dot{\xi}_1 - \dot{\xi}_2 \Rightarrow B_1 = B_5 + \dot{\xi}_1 + \dot{\xi}_2 = 2 + 121 + 31 = 154$

2nd $P: P_8 = \dot{\xi}_1 - \dot{\xi}_2 \Rightarrow \dot{\xi}_1 = P_8 + \dot{\xi}_2 = 90 + 31 = 121$

1st $Q: Q_5 + Q_8 = \dot{\xi}_2 \Rightarrow \dot{\xi}_2 = 30 + 1 = 31$

Separator 2 Balance

$P_7 = P_8 = 90$

$Q_7 = Q_8 = 1$

$B_7 = B_9$

Separator 1 Balance

$A_3 = A_4$

$B_3 = B_4 + B_7$

$P_3 = P_7 = 90$

$Q_3 = Q_4 + Q_7$

Fractional Conversions

$$\frac{A_2 - A_3}{A_2} = 0.1 = \frac{2\dot{\xi}_1}{A_2} \overset{3rd}{\Rightarrow} A_2 = \frac{2\dot{\xi}_1}{0.1} = \frac{2(121)}{0.1} = 2420$$

$$\frac{B_2 - B_3}{B_2} = 0.25 = \frac{\dot{\xi}_1 + \dot{\xi}_2}{B_2} \overset{4th}{\Rightarrow} B_2 = \frac{\dot{\xi}_1 + \dot{\xi}_2}{0.25} = \frac{121 + 31}{0.25} = 608$$

Reactor Balance

1st $A_3 = A_2 - 2\dot{\xi}_1 \overset{5th}{=} 2420 - 2(121) = 2178$

2nd $B_3 = B_2 - \dot{\xi}_1 - \dot{\xi}_2 \overset{6th}{=} 608 - 121 - 31 = 456$

$P_3 = \dot{\xi}_1 - \dot{\xi}_2 \overset{7th}{=} 121 - 31 = 90$ (already found)

(SAVE) $Q_3 = Q_2 + \dot{\xi}_2$

Splitter Balance

$A_4 = A_5 + A_6$

$B_4 = B_5 + B_6$

5th $Q_4 = Q_5 + Q_6$

$Q_6 = Q_4 - Q_5 = 6534 - 30 = 6504$

split

$$\frac{A_5}{A_4} = \frac{B_5}{B_4} \overset{1st}{\Rightarrow} B_4 = B_5\left(\frac{A_4}{A_5}\right) = 2\left(\frac{2178}{10}\right) = 435.6$$

likewise 2nd $Q_4 = Q_5\left(\frac{A_4}{A_5}\right) = 30\left(\frac{2178}{10}\right) = 6534$

3rd $A_6 = A_4 - A_5 = 2178 - 10 = 2168$

4th $B_6 = B_4 - B_5 = 435.6 - 2 = 433.6$

Mixer Balance

$B_6 + B_1 + B_9 = B_2 \Rightarrow B_9 = B_2 - B_1 - B_6 = 608 - 154 - 433.4 = 20.4$

$Q_2 = Q_6 = 6504$

Recall Rx eqn

$Q_3 = Q_2 + \dot{\xi}_2 = 6504 + 31 = 6535$

or sep 1 $Q_3 = Q_4 + Q_7 = 6534 + 1 = \underline{6535}$

	1	2	3	4	5	6	7	8	9
A	252	2420	2178	2178	10	2168	—	—	—
B	154	608	~~456~~	435.6	2	433.4	20.4	—	20.4
P	—	—	90	—	—	—	90	90	—
Q	—	6504	6535	6534	30	6504	1	1	—

(b) Selectivity

$$\frac{\text{moles of } B \to P}{\text{moles B consumed}} \overset{SP}{=} \frac{\left[(A_2 - A_3)\frac{\nu_B}{\nu_A} - (Q_3 - Q_2)\right]}{B_2 - B_3} = \frac{\dot{\xi}_1 - \dot{\xi}_2}{\dot{\xi}_1 + \dot{\xi}_2} = \frac{90}{152} = \underline{\underline{0.592}} \text{ single-pass}$$

"Produced P"

$$\overset{OV}{=} \frac{P_8}{B_1} = \frac{90}{152} = 0.\underline{\underline{592}} \text{ overall}$$

The same

Yield

$$\frac{\text{moles of P produced}}{\text{moles B fed}} \overset{SP}{=} \frac{90}{608} = \underline{\underline{0.148}} \text{ single-pass}$$

$$\overset{OV}{=} \frac{90}{154} = 0.\underline{584} \text{ overall}$$

differ

Recycle doesn't affect selectivity, but does improve yield.